

Introduction to QGIS

exploring their features and performing map composition

Laboratory Exercise for the Course

Geographic Information System and Remote Sensing

B Tech in AI
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A. Download and install: <https://qgis.org/en/site/forusers/download.html>

B. Basic QGIS - PART 1. This file demonstrates loading .shp and .csv files.

- Sample CSV file:
 - <https://github.com/bidur/GPSProbeGeneration/blob/master/input/original.csv>
- Shapefile:
 - <https://github.com/bidur/nepal>
- Geotiff
 - <https://github.com/bidur/nepal>

C. Load CSV data and explore Coordinate System

- i. Load CSV as Points
Layer → Add Layer → Add Delimited Text Layer
- ii. Load data in Geographic Coordinate System (*WGS 84* → *Coordinates are in Latitude & Longitude (degrees)*) and display on map
In dialog box:
 - File: Select your CSV
 - Geometry Definition:
Point coordinates
 - X Field → **Longitude**
 - Y Field → **Latitude**
 - Geometry CRS →
Select WGS 84 (EPSG:4326)
- iii. Examine Coordinate System
Check Layer CRS
 - Right-click layer → Properties
 - Go to SourceYou can see “CRS: EPSG:4326 (WGS 84)”
- iv. Reproject Layer(i.e. Convert to GCS to Projected Coordinate System)
Right-click CSV layer
→ **Export** → **Save Features As**

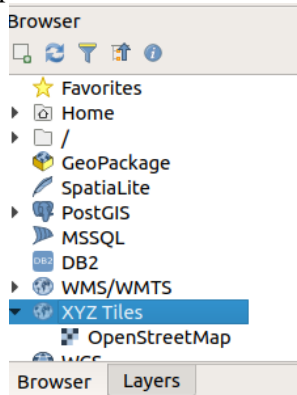
Select UTM CRS

- Format: Shapefile
- CRS: Search:
EPSG:3857 (WGS 84 / Pseudo-Mercator)
- Save as: new_crs_points.shp

Note: Google Maps primarily uses **EPSG:3857** (WGS 84 / Pseudo-Mercator)

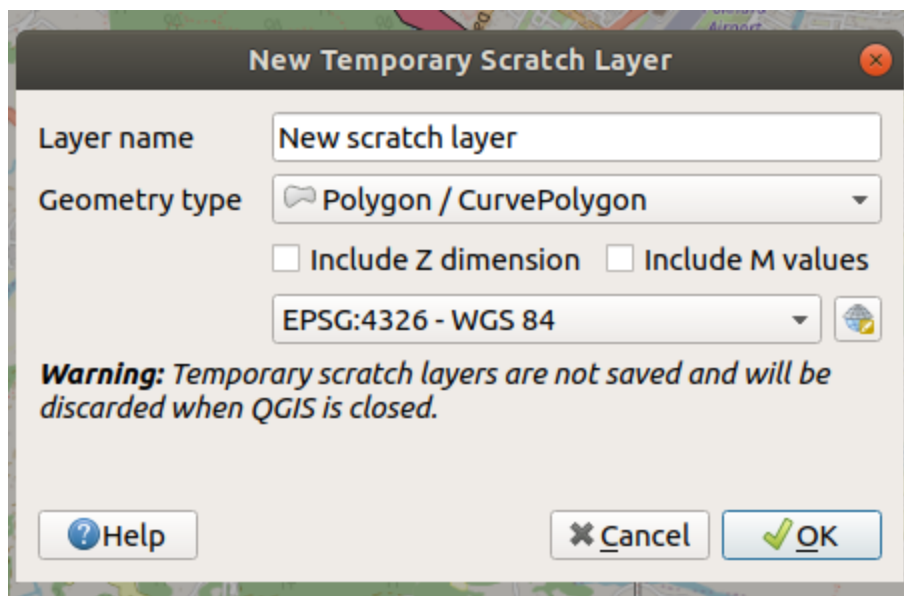
D. Make a Map highlighting Lakes in Pokhara (Phewa, Begnas, etc)

- i. First, add basemap. Go to browser panel and choose XYZ tiles and select OpenStreetMap.



OpenStreetMap is added to the canvas.

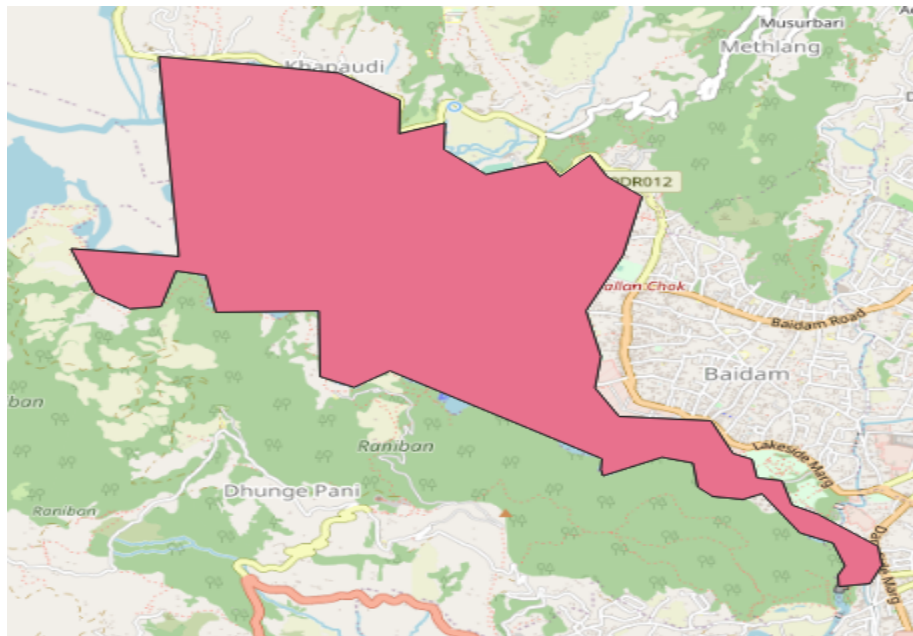
- i. Plugins can also be installed. For example: Plugins → Manage and Install Plugins → QuickMapServices
- ii. Create a new layer:
 - i. For creating new layer go to **layer** in the menu toolbar. Select New temporary scratch layer. Fill the layer name and select the geometry of the layer you want.



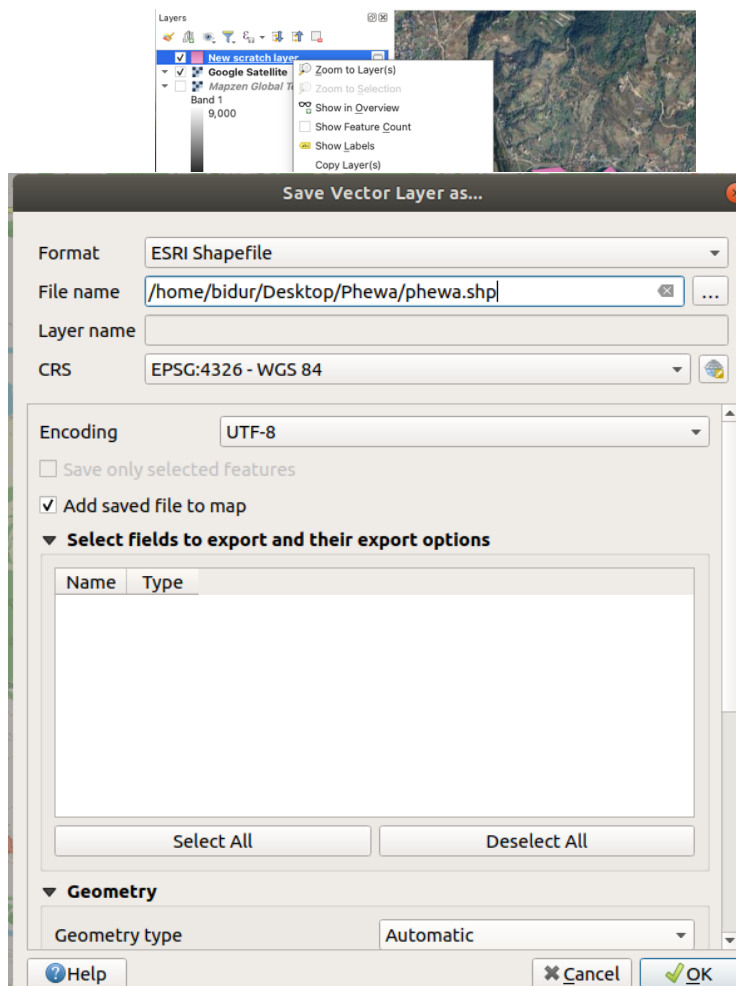
- iii. Select Add polygon feature from digitizing toolbar.



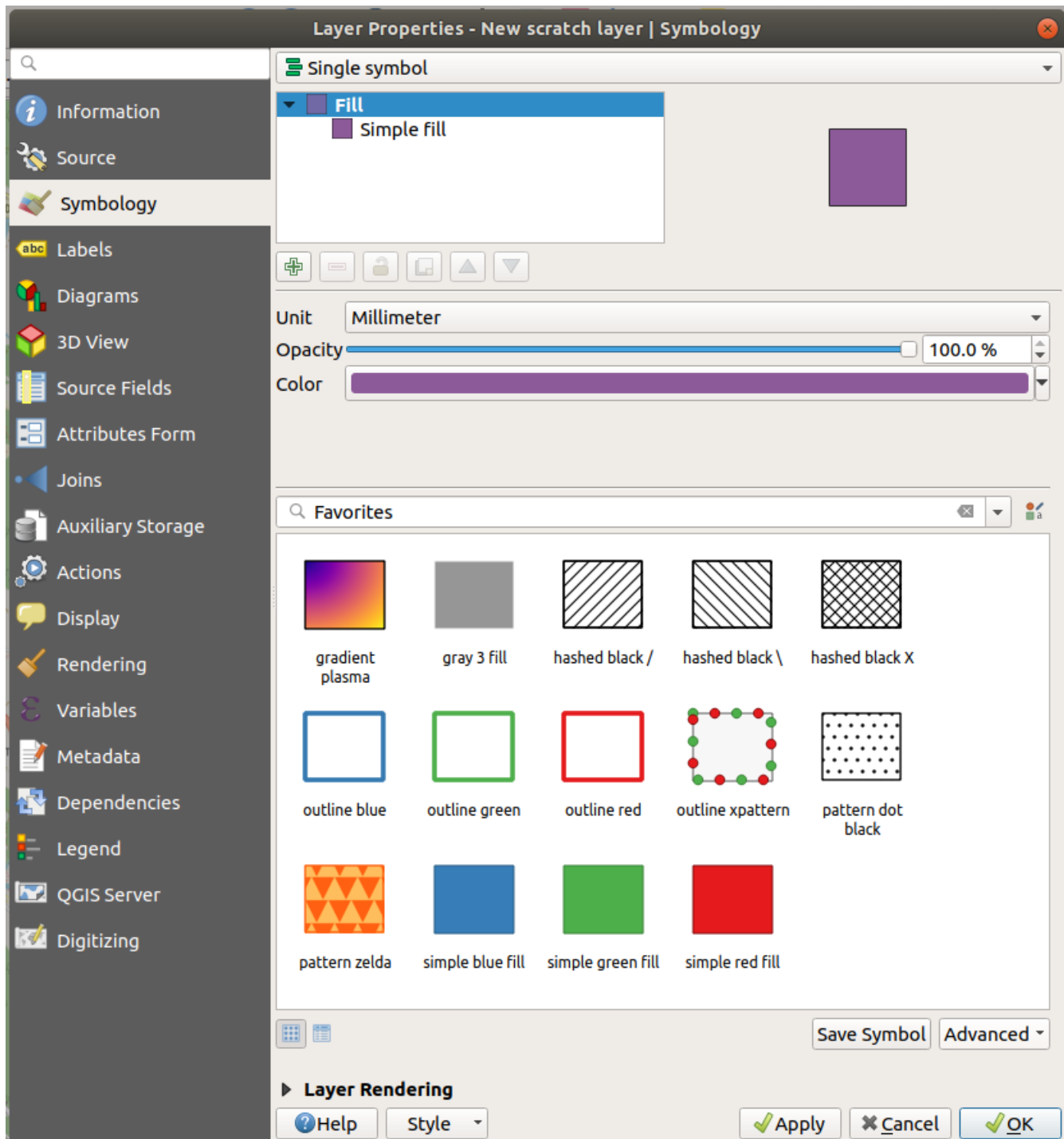
iv. Now Add Polygon Feature, and show lakes. Phewa lake is shown below:



- v. Make use of toggle editing by clicking toggle icon.
- vi. After completing all lakes, Save layer by right clicking on the layer. Tap on **Export** and click **Save Features As**.

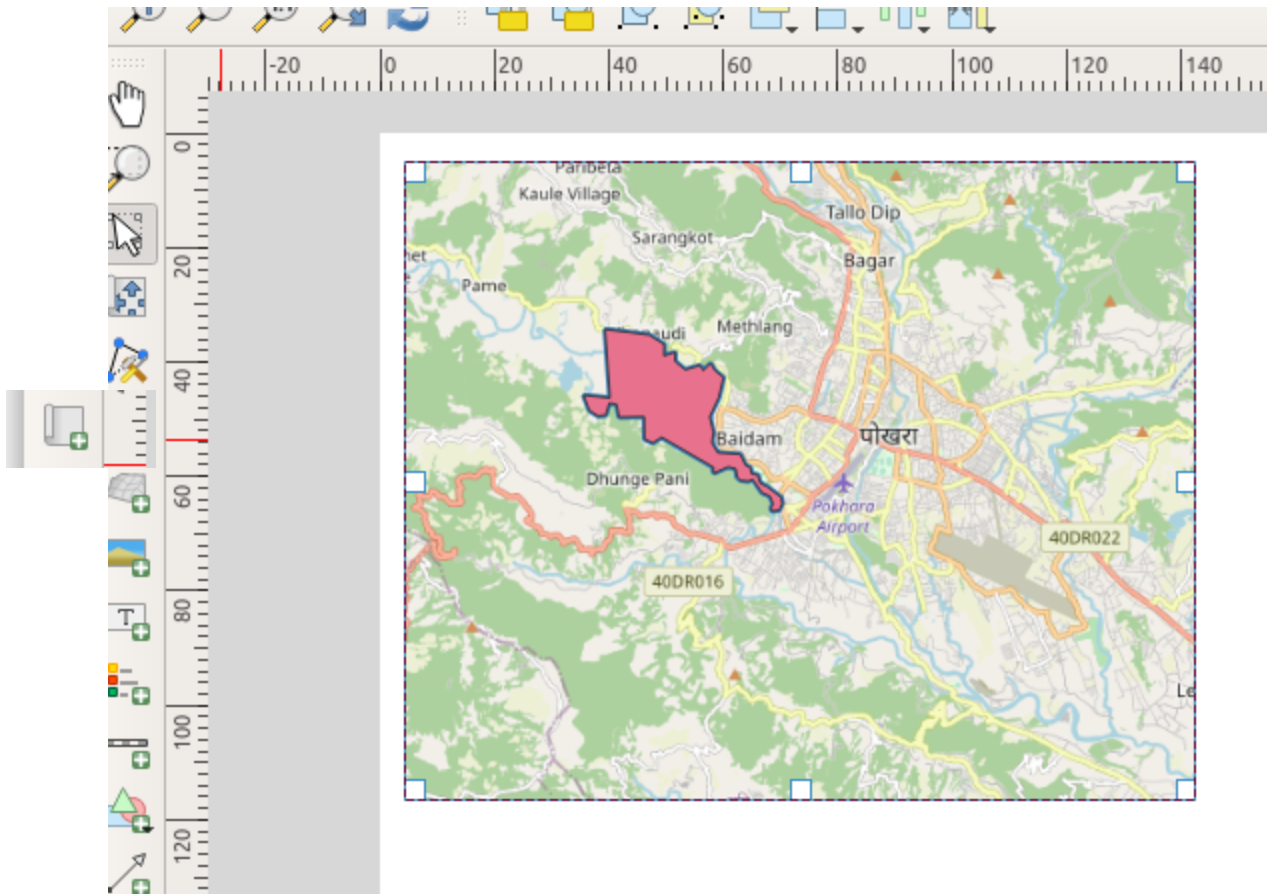


- vii. Right Click “Layer” and explore Properties to check the layer information and to change/add other stuffs like symbol, labels, etc.



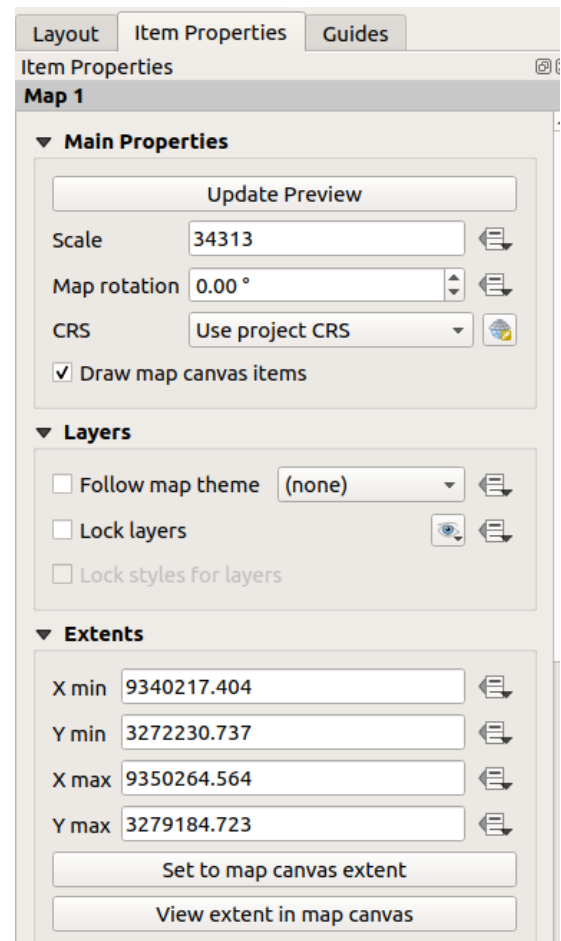
Printing Map

1. Project → New Print Layout
2. Click Add Map to Layout . Click mouse pointer in blank region. Then the layers will render in the screen.



3.

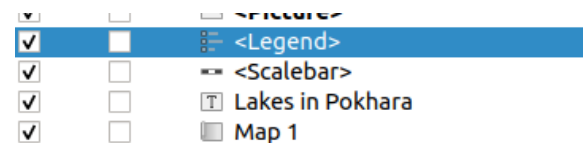
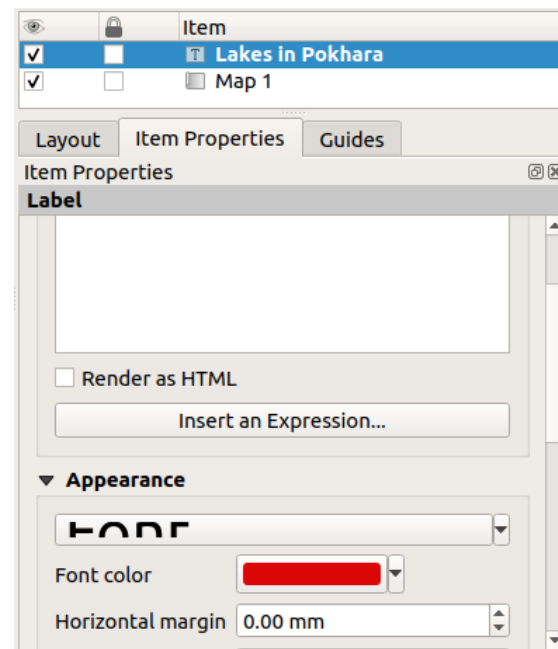
You can change the size and scale of the map using different buttons and commands. Explore the User Interface buttons.



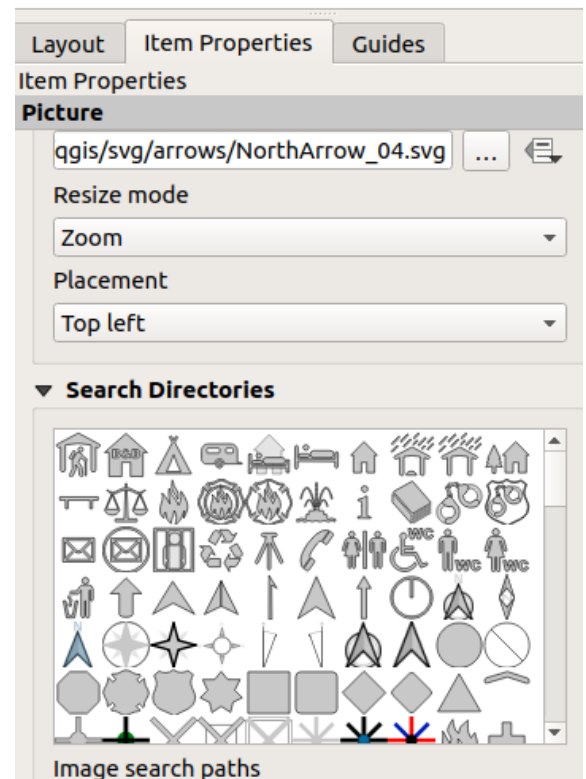
4. Any changes done in the main layers will be reflected in Print Layout interface. Try clicking the refresh button to enforce it.
5. Add Map Title by clicking the button.



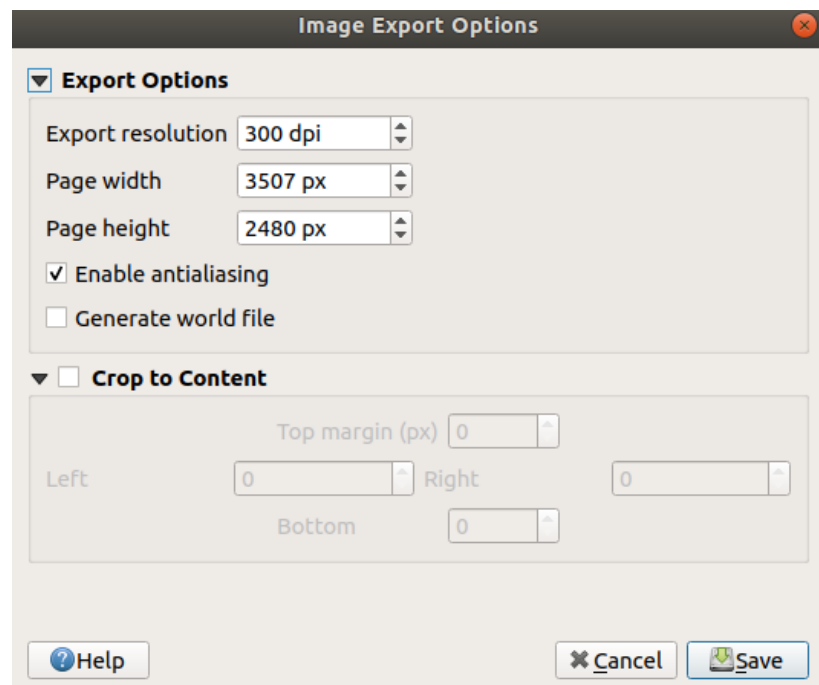
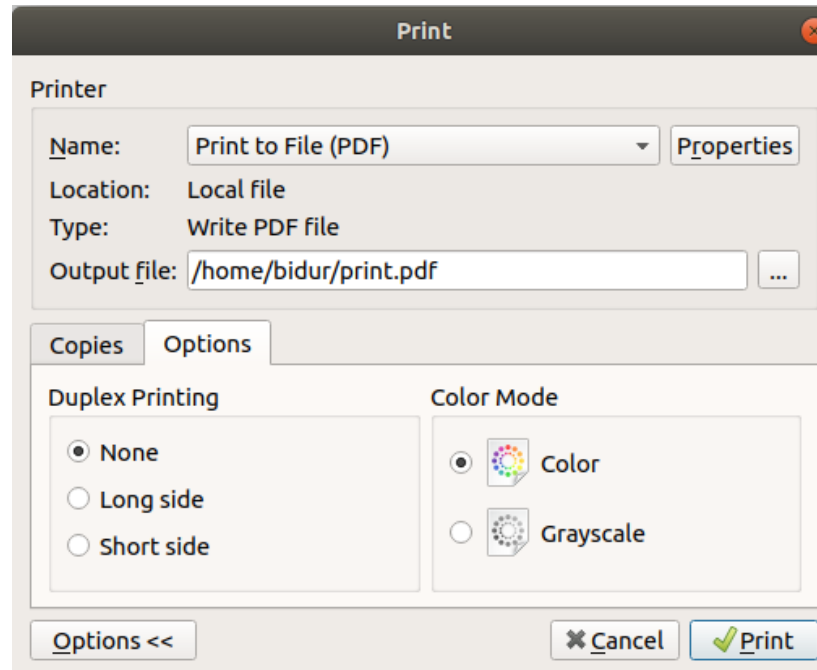
- Update the Title properties like font, color, etc from the properties window.



- Similarly Add ScaleBar by clicking the button in the UI.
- Similarly, add Legend. Uncheck “Auto Update” in the Legend Item Properties , so that you can add/remove the legend you wish.
- Similarly, add Image (for adding Map direction). Search directories and add the Arrow Image.



10. You can now print the map. Explore different options and properties before printing.
11. You can also, export the map as image or pdf.



References:

https://docs.qgis.org/2.8/en/docs/training_manual/basic_map/index.html

<https://www.igismap.com/qgis-3-4-4-tutorial-creating-editing-and-mapping-in-gis/>

https://www.youtube.com/watch?v=YCDBjl_xzAY